Project Proposal

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Research Report

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ABSTRACT

Finding a phone number that appeals to the customer could be a challenging and time-consuming task. The number of available phone numbers could be overwhelmingly large. Furthermore, the concept of what is easy to remember can be quite subjective from customer to customer. Each person has their own combinations of numbers which carry certain personal meaning, like dates, lucky numbers. Because of these two problems, finding the optimal number could take a lot of time.

This project aims to develop a system that automates this process. It identifies and ranks numbers, also license plates, which fall into a category of memorable by matching certain rules to the data available in a dataset. The rules represent certain numerical patterns such as repetition, sequences, symmetry, and palindromes.

This system uses a modular design where each rule functions as an independent pattern detector that contributes to the overall memorability score computation. This way the system is easy to test and implement incrementally.

TABLE OF CONTENTS

Authorship Declaration ii
Acknowledgements iii
Abstract iv
Table of Contents v
1 Introduction 1
2 Background and Motivation 1
2.1 Aim and Objectives 2
2.2 Organisation 2
2.3 Background Research 2
2.3.1 Human memory and numeric patterns 3
2.3.2 Numeric memorability 3
2.3.3 Alphanumeric pattern memorability 4
2.3.4 Algorithms and system architecture .....**Error! Bookmark not defined.**
2.3.5 Natural language processing 4
2.3.6 Similarity metrics and search 5
2.3.7 Rule-based weighted ranking system 6
2.3.8 Temporal-Difference learning 6
2.3.9 Modular multi-agent system 7
3 Methodology and work plan 7
3.1 System workflow for phone number system 8
3.2 Methods 8
3.3 Ranking rules and scoring 9
3.4 Requirements 10
3.4.1 Functional requirements 10
3.4.2 Non-functional requirements 10
3.5 Research goals 11
3.6 Research methods 11
3.7 Stakeholders and use cases 11
3.8 Tasks 12

3.9 Evaluation strategy 12
3.10 Scalability.....12
4 Feasibility 13
4.1 Technical feasibility.....13
4.2 Resource and time feasibility 13
5 Conclusions.....14
5.1 Motivation and goals 14
5.2 Proposed work.....14
References 15
A. Project Management 16
A.1. Gantt Chart.....18
A.2. Risk Analysis.....19
A.4. Rules 20
A.5. Diagrams 21
A.1. Ples.....23
B. Generative AI 23

1 Introduction

Automating small tasks that can be time-consuming is a smart decision for any business, and a cost-effective solution. A distinctive phone number or a licence plate is not only useful for a client, making it easier to remember but also for a business to be recognisable and look more professional. However, the process of identifying an ideal number manually is a complex task. Each new number that is picked by a customer can be unique, depending on their personal preferences and needs, making it challenging to select the optimal choice.

Certain numerical patterns can be classed as easy to remember. They tend to adhere to simple rules, such as symmetry, repetition, and sequences. Number combinations such as 50505050 or 123321 stand out from other types of patterns, for example, 481632 (geometric sequence). That is why it is important to implement a ranking system that ensures that selections are optimal.

This project proposes a rule-based modular design system that has predetermined rules, which acts as independent pattern detectors. The combination of such modules evaluates appearing patterns in the numbers structure and contributes to the overall ranking score, which serves as the numbers' evaluation of how memorable the combination is. Optionally, temporal differences learning could be to update the weight of each rule based on user feedback. This design allows the system to be responsive to each user's preferences using dynamic reinforcement.

In this project terms like pattern detector, module, and agent are used interchangeably. The project proposes the artificial intelligence concepts which are optional, so the correct terms for such system components depend on the ai concept inclusion in the system. Also, the report is using term memorability to keep the consistency throughout the report, but that also include the level of interestingness of the pattern.

2 Background and Motivation

The motivation for this project is to develop a system that businesses can use to optimise their daily tasks, making it more profitable and timesaving. The purpose of recognising memorable phone numbers for a business can be influential to advertising and brand consistency. Similarly, individual customers often prefer numbers that are easier to remember or have significant personal value, including certain combinations of numbers resembling important dates, lucky, or angel number formations. Choosing the optimal number from a vast dataset of available ones can be an impossible task without the help of such an automated system.

This project is using principles of pattern detection, and natural language processing. Its modular design enables each rule acts as a separate process. The temporal difference learning is considered as an additional feature to provide adaptability to the system and

could enable the system to update its logic of what is memorable according to the user, with limited feedback.

2.1 Aim and Objectives

The aim of this project is to design a system that can identify and rank memorable phone numbers, or car plates using predetermined rules.
The objectives of this project to achieve this aim are listed below.

1.	Designing a modular rule-based system which can detect patterns in numeric, alphabetic or alphanumeric data.
2.	To integrate lightweight natural language processing methods such as approximate string matching, and similarity search, to determine if they can adequately contribute to memorability score when it comes to number plates.
3.	Develop a ranking system where each pattern detector contributes to an overall memorability score optionally using temporal difference learning on user feedback.

2.2 Organisation

This document is structured into six main sections to define the problem and main objectives, feasibility, and a review of the research methodology required for the completion of the system.

- Section 1 defines the motivation, aims and objectives for the creation of such a system.
- Section 2 provides background research analysis. The first part of this section is cognitive research for defining what is memorable, what the preferences are to refine the vague term of memorability, and to make an educated decision about the rule ranking. Part two of this section overviews system design and algorithmic methods.
- Section 3 describes the system workflow, methods, design, modular architecture, requirements, research methods, tasks, evaluation strategy. It recognises project stakeholders and use cases are overviewed in this section.
- Section 4 is dedicated to project feasibility. It confirms that from technical, resource and time perspectives, the project is feasible.
- Section 5 includes conclusions about this project.

2.3 Background Research

The research that is required for this project consists of cognitive psychology about numeric and alphabetic patterns that are proven to be memorable, modular multi-agent architecture, and natural language processing concepts. This section reviews foundation needed for developing pattern ranking logic.

2.3.1 Human memory and numeric patterns

The important part of this project is to define the specific patterns that are more memorable to people. This includes not only numeric sequencies, but also alphabetic combinations. Also, not every pattern is equally appealing, which raises the problem of the hierarchy of patterns. The decisions on ranking the patterns by their memorability must be made.

2.3.2 Numeric memorability

Research shows that humans prefer structured, meaningful or in some cases familiar patterns in random alphanumerical combinations. The article on memory of numeric security codes of Martin Ramussen and Floyd W Rudmin shows that few of the top strategies for remembering the number sequences are "The next most common mnemonic strategy, reported by 54%, was to remember the pattern the code makes when entered on the keypad. The strategy of using security codes from other contexts was reported by 40%, and choosing a code with a personal meaning was reported by 38%" (Rasmussen and Rudmin, 2010). This is quite an important observation for this project, considering what strategies people use to make it easier to remember number sequences. These results suggest that 54% of their study participants tend to rely on the visual placement of the number sequence in the keypad. Some easy-to-remember sequences following such a strategy could be straight line patterns such as 123, 963, 357. Also, personal meaning is an extremely helpful strategy for recalling numerical patterns, which emphasises the need for user input and choice functionality in the system. Furthermore, it shows that participants relied more on contextual cues rather than randomness.

An article written by Cowan. N and Hardman. K. O. shows that "trains were generally helpful to recall, especially when they fell within a triad but to some extent even when they crossed triad boundaries" (Cowan and Hardman, 2021, p, 744). It supports the rule about repeated number sequences being a top priority when it comes to numerical pattern recalls. Another observation has been done on the repetition of numbers, which indicated high memorability. "There was some evidence that lists with 4 unique digits produced better recall than lists with less repetition and therefore more unique digits" (Cowan and Hardman, 2021, p. 752). It indicates that the least amount of unique numbers improves memorability. Also, this study shows how grouping can emphasise certain numerical patterns, leading to quicker evaluation of appealing combinations.

Furthermore, the Jones, Mari Riess, and June J. Baird's article explores the recall of three numerical patterns, such as symmetrical patterns, arithmetic sequences, and random patterns. "In general, it appears that symmetrical patterns are indeed simpler than patterns with other types of structure… In both serial and free recall modes, sequences with higher-order nominal redundancies were more accurately reconstructed than ones with arithmetic rules or lower-order structure alone" (Jones and Baird, 1979, p. 640).

The findings show that symmetrical patterns take the first place as memorable patterns, followed by simple sequences.

2.3.3 Alphanumeric pattern memorability

Similar concept applies to license plates, which adheres to all the patterns mentioned before, although contains additional complexity. The addition of alphabetic characters to the number combinations raises the problem of applying the natural language processing concept to the system to recognise linguistic and phonetic patterns. Semantic meaning containing letter strings has a superiority when it comes to memorability in comparison to simple patterns adhering to numbers such as symmetry, repetition, or sequencies.

In visual cognition symmetry and shape similarity improves the memorability. So certain letters or letter combinations would be recognized faster. For example, AHA, TOT in comparison to DOD or ERE.

Probably the most important pattern for alphabetical combinations is relatability with semantically meaningful words. Either the letter combination is exact match with a word or having numbers which either phonetically represents parts of words, or numbers physical resemblance to letters. Such combination is still highly recognizable and memorable.

2.3.4 Natural language processing

Since number plate memorability depends on recognizing meaningful words, or phonetic interpretation of numbers, the system must use natural language processing components, which can detect exact words, or distorted versions of words with certain numeric digits representing letters.

Although memorability for alphabetical structures does obey the same rules as numeric patterns, nonetheless semantic meaning is highly important. The combinations of letters that can represent words will require number to letter transformations, approximate matching, also phonetic number and letter values taken in consideration.

To handle such complexity, the system will incorporate character level natural language processing to analyse sequencies letter by letter. This is beneficial to such system, where the number of characters is highly limited, and they mostly contain resemblance to, or shortened versions of real words. With this approach system can compare how closely singular strings which consists of alphabetic and numeric combination resemble actual words, recognising linguistic patterns.

The character level similarity search techniques are applicable to short text analysis, where text distortions can be present. Hybrid lightweight models combining character level and similarity metrics proposed by Wibowo in Lightweight String Similarity Approaches for Duplicate Detection in Academic Titles are closely related to system design proposed in this report. Such models ensure that even alphanumerical combinations can be interpreted as deviation from words with semantic meaning.

The exact dictionary match alone, can miss on close match cases, which can still be appealing to the user. The letter combination can closely resemble words with extra letters or numbers interjecting the combination. For example, CAAR, MOO7N. In this case the approximate string matching would enable the system to detect potential suggestions to the user even with slight deviations from meaningful words.

Another challenge for such system is the phonetic pronunciation of the number. This can be used to resemble a word or even few words with least number of characters. For example: 4ever, me2, 2ring. Many number plates use leet transformations, where numbers are matched to the letters depending on their physical appearance resembling those letters. For example, 8-B, 1-I or L, 5-S. Such substitutions can be treated as character level variation as described in approximate string matching. This enables the wider potential number of car plates to be considered as memorable. Although "the problem of approximate string matching as that of finding the text positions that match a pattern with up to k errors." (Navarro, 2001) Therefore, the system must not consider cases, where the dataset record has too big of a difference from the given example string. Nevertheless, this concept ensures that close match with small deviations cases are not overlooked by the system. In this case approximate string matching is beneficial.

For each of these functionalities such as exact approximate dictionary match, phonetic mapping, and leet substitution, the system will generate the memorability score that indicates the presence of such pattern and its strength. In this case, the appropriate weight of each concept in license plate pattern detection is as important just as phone number memorability weighting. For each type of pattern, the normalized scored must be given, which then can be combined into one memorability score.

The natural language processing enables the system to treat the number plates, not only by their numeric pattern presence, but also by phonetic and semantic meaning.

2.3.5 Similarity metrics and search

Similarity search in this system allows user to get suggestions of similar patterns to their preference. The problem of finding such combinations in the dataset may be problematic, for no exact match being present in the dataset. To overcome this problem the system will use different distance and similarity metrics to suggest the closest matches. These may include vector-based similarity, and character level distance metrics.

2.3.6 Rule-based weighted ranking system

The integration of multiple rules which include numeric, alphabetic and sematic types, creates a need of a unified and sophisticated ranking system with predetermined rules.

The systems general logic consists of rule-based modules responsible for detecting the rules presence in analysed number or car plate and combined into general memorability score. Each module is responsible for singular type of pattern detection, such as symmetry, word similarity, sequencies. This type of system closely resembles the hybrid lightweight frameworks described in Wibowo work. Independent components contribute to united similarity score.

Alphabetic patten detection uses natural language processing methods such as character approximate string matching, dictionary lookup, leet transformations. Dictionary word detection is one of the most important methods to be used for alphabetical pattern detection. The random letter combination which has a semantic meaning is much easier to recall. To detect such letter combinations, the leet speech type of mapping may be applied to potential letter sequencies which can represent the resemblance of certain numbers to letters. For example, see the rules attached in section Appendix . If a word is detected, then the pattern detector dedicates appropriate score, which is suggestive of the strength.

Because each pattern is a modular component, extra rules can be added without modification of the overall system. The separate pattern detectors contribute to the memorability score.

2.3.7 Temporal-Difference learning

Temporal difference learning is considered as an optional feature to upgrade the system into learning from user feedback. The user will most likely make from 3 to 5 asks for new, different number recommendations, before just picking the most appealing one. The more users interact with the system, the easier it is for the system to tailor suggestions, adhering to the users' preference logic. To make the system optimal, the learning functionality must be quick to adapt. For that reason, the learning method to be used for this type of system must be model-free to make adaptive weight adjustments without predefined assumptions of user actions.

In the Sutton and Barto book "Reinforcement learning: an introduction," it is defined as "TD learning is a combination of Monte Carlo ideas and dynamic programming (DP) ideas. Like Monte Carlo methods, TD methods can learn directly from raw experience without a model of the environment's dynamics. Like DP, TD methods update estimates based in part on other learned estimates, without waiting for a final outcome" (Sutton and Barto, 1998, p. 120). This combination is perfect for this project because it allows each user's choice to be treated as a learning event. This way, the system is dynamically

reinforcing its decision on what the memorable number is. Such system is delivering more accurate personalised recommendations with each user's choice.

When personal preferences of what is memorable are considered, the temporal difference learning process would enable the adjustment of the rule hierarchy by incrementally adjusting their weights. This could make the system highly responsive and personalised.

2.3.8 Modular multi-agent system

The system's modular design allows for treating each rule as a separate agent, maintaining its own weight and TD updates. Learning agents collectively determine the overall memorability score by their independent contributions, which are based on individual rules. A box diagram may be seen attached in Appendix.

This type of structure, where multiple agents collaborate towards a common goal, is similar to Kurt-Nelson and Redish's proposed multi-agent architecture. “Each µAgent maintains an independent hypothesis about the state of the world, and a separate value-estimate of taking actions within that hypothesized state. The overall agent thus instantiates a flexible representation of an evolving world-state” (Kurt-Nelson and Redish, 2009, p. 3). This way system achieves modularity, each improves the ranking system logic in comparison to a singular agent for making an overall estimation on the possible weight updates rather than a single global model.

Furthermore, this structure aligns with the multi-agent architecture proposed by Russell and Norvig “Cooperative game theory considers settings in which agents can make binding agreements to form coalitions to cooperate. Solution concepts in cooperative game attempt to formulate which coalitions are stable (the core) and how to fairly divide the value that a coalition obtains (the Shapley value).” (by Russell and Norvig, 2021) All the agent contributions, and their differently weighted combination, make the final memorability score.

By maintaining modularity, the system ensures to be extensible, explainable, adaptable. This supports isolated testing, explainability and scalability of the system.

3 Methodology and work plan

In this section the workplan for this project is outlined and explanation provided of how the researched methods will be used within the system. The proposed system design incorporates the following concepts: rule-based ranking, character level natural language processing, approximate string-matching technique and possibly adaptive learning. This ensures that the system can adequately evaluate the memorability of each item in the dataset and give appropriate suggestions to the user. With these methods, the system will be capable of detecting exact matches and closely related with small deviation matches.

3.1 System workflow for phone number system

The system is designed as a modular workflow and consists of atomic operations. Each operation has a specific ranking and pattern-recognition task. The following workflow is depicted in a cross functional diagram which is attached to Appendix. To start with, the system must load the available phone numbers from a given dataset and calculate a memorability score. The system will calculate the scores for each available phone number using a weighted combination of predefined memorability rules, such as symmetry, repetition, sequences, keyboard patterns, and unique digit count. This way, the user gets the first 5 suggestions of phone numbers with the very best rank.
The user then has three options.

1. Select one of the five suggested phone numbers as theirs.
2. Request similar numbers to one of the five suggested numbers as an example combination. In this case, the cosine similarity method could be applied to find similar number patterns.
3. Input a sequence of numbers with a desirable pattern.

In case the user decides to input their own sequence of numbers, as a priority, the system must try to find an exact match of the pattern present in the available numbers dataset using string comparison. If the exact match does not exist containing the sequence, then the system uses Manhattan distance to determine the closest distance between the suggested and available numbers. Finally, the system makes use of the memorability score to make sure that the number is not only the best match to the user's input but also is one of the highest scoring suggestions for the memorability score, applying weighted ranking.

Optionally, if temporal difference learning is implemented, the system could gradually adjust the weights with each user interaction, to improve the personalised suggestions. The TD error will be calculated to determine how much the expected and observed memorability score differ. Then the TD error will be used by micro agents in the calculation of individual weight updates. The agent takes into consideration of how much the rule contributed to the number, which is set by the score of that feature. The system will use predefined weights if such functionality is not implemented. Finally, the system combines agents' calculations to make a new personalised memorability score and gives the user the best ranking number.

3.2 Methods

To achieve the objectives, the following methods will be used: natural language processing, rule-based ranking, modularity, multi-agent system, and optionally temporal difference learning.

Each rule is an atomic operation that analyses one aspect of the phone number or a license plate structure, adhering to a modular design. This will ensure an easier testing

procedure and data integrity. The rules check the symmetry, repetition, sequences, keyboard patterns, and unique digit count. Each outcome is normalised to a number from 0 to 1 for compatibility. Then the features are combined using a weighted score. This will produce the estimate for a memorability final score for a phone number or a number plate.

The natural language processing methods perform character level analysis of alphanumerical sequencies. The methods related to it will perform string matching, dictionary search, and leet transformations. This enables the system to detect meaningful words despite of alphanumeric combinations containing deviations.

The ranking mechanism in this system could optionally use temporal difference learning to incrementally adjust the weights for each of the rules, making the suggestions more personalised. Without this feature, the system will use predefined weights. Because each rule acts as a separate meta or micro agent, the weight adjustments will be calculated independently and combined by the system into an overall adjusted memorability score.

3.3 Ranking rules and scoring

To rank the numbers as memorable, as mentioned in previous sections, it is important to distinguish the exact rules and hierarchy. At the beginning, the system will not have the users' preferences for what is memorable to them, which is why the ranking is important. If temporal difference learning is implanted, the further in the process of picking the phone number or a car plate, users' picks will implement the system's decisions of what is memorable in this specific case.

The exact rules are provided in Appendix, which are going to be used for initiating the ranking. They are organised into sections of numeric patterns, alphabetical patterns, combination of numeric, and alphabetic, and semantically meaningfulness. Taking into consideration that the cognitive studies analysed in the research section for number combination memorability are not a single study with further defined rules ranked systematically against each other, there is no initial hierarchy of the rules. Memorability may be quite subjective, so ideally suggestions will not be assumed to be the most appealing ones for the user. The user's choices will implement the system to find the most memorable case by comparison and weight. The final score will be a combination of preset logic rules and model user preferences. Each user interaction will feed further model learning.

As micro agents evaluates each numeric or alphabetic feature present in the given phone number or car plate, eventually system will combine numeric pattern scores and natural language processing-based ranking into a single memorability score.

3.4 Requirements

The Requirements for this system are written using the Mosco method. They are ordered into the requirements that must be implemented; otherwise, the system would not be developed to the point where the main objectives could be tested. The requirements ranked as should have defined extra functionalities that such a system would benefit from. The requirements with could have ranking defines the requirements which may be implemented if there is enough time. The requirements ranked as won't are mentioned only for the future development of the project but will not going to be implemented at this stage of the project development.

3.4.1 Functional requirements

FR 1	The system must load a dataset of available phone numbers.
FR 2	The system must detect numerical patterns based on predetermined rules.
FR 3	The system must detect linguistic patterns in number plates using NLP methods such as:
FR 3-1	Dictionary matching.
FR 3-2	Approximate string matching.
FR 3-3	Phonetic similarity.
FR 3-4	Leet transformation.
FR 4	The system must successfully calculate a memorability score.
FR 5	The system must display top ranked suggestions to the user.
FR 6	The system must manage modular rule components, which calculate weight updates that contribute to overall memorability score.
FR 7	The system should enable the user to select one of the suggested phone numbers as the best one.
FR 8	The system should enable user to request for similar suggestions to desired numeric patterns.
FR 9	The system should allow user to enter a custom numeric or alphanumeric pattern.
FR 10	The system should store previous users' choices and inputs.
FR 11	The system could update the ranking weights dynamically based on user feedback using temporal difference learning.
FR 12	The system could provide explanations for the ranking decisions.
FR 13	The system will not work on external dataset integration.
FR 14	The system will not support user authentication.

3.4.2 Non-functional requirements

NFR 1	The system must be able to read at least 1000 phone numbers or car plates from a dataset.
NFR 2	The system must be able to rank at least 1000 phone numbers in under 5 seconds.

NFR 3	The system must show adequate suggestions of phone numbers or car plates, clearly adhering to memorability rules.
NFR 4	The system should be able to work with larger datasets, for example 10 000 phone numbers, performing under 10 seconds.
NFR 5	The system should provide error messages.
NFR 6	The system could allow the user to select a preference for which rule of memorability adheres the most.
NFR 7	The system could show the rationality analysis of why the numbers were selected and suggest to the user.
NFR 8	The system won’t collect any personal data.
NFR 9	The system won't have a mobile version application.

3.5 Research goals

The research aims to answer the following questions.

Can a modular rule-based architecture effectively separate alphabetic, and numeric pattern detection into independent pattern detectors? (Ob1)
Can lightweight natural language processing methods be successfully integrated to evaluate the memorability of alphanumeric number plates? (Ob2)
Can a rule-based weighted system optionally using the TD learning method create a reliable memorability ranking? (OB3)

3.6 Research methods

For the research, which was needed for this project, the Heriot-Watt University Discovery website was used. Only the sources that support the method decision was used. The deductive reasoning method was used with a top-down approach. After defining the main deliverables of the system, such as learning from user data, finding similarity, and defining the memorability, I have started narrowing down by methods that are not applicable for this project or do not align with the predefined workflow.

After main deliverables were defined, the research continued by narrowing down choices for methods. Some techniques, such as supervised learning, clustering were researched but named as unsuitable for such project due to not aligning well with modularity, small user feedback, or timeframe. The methods that have been chosen, aligned well with system workflow. This deducted reasoning approach ensures that the project is technically feasible.

3.7 Stakeholders and use cases

This system's main stakeholder is the telecommunications business. A system as such would automate the process of picking the phone numbers, optimising the finding of the most distinctive, appealing, and memorable phone number for the user. The use case for

this type of system could be in the shop while working with clients in real-time, also integrated into an online application for ordering a new phone number online, making the process more accessible to the user.

3.8 Tasks

This project is following Agile methodology, which perfectly aligns with the systems modular design. Iterative implementation and testing align well with this methodology, so each task can be set as a 2-week sprints for first semester and 1-week sprints for the second semester. There will be concurrent work related to implementation and testing of the system, also the dissertation writing. The project plan mostly separated in weekly sprint style can be seen attached in Appendix.

3.9 Evaluation strategy

The project evaluation will consist of two parts: computational evaluation and functional behaviour validation.

In computational evaluation, the main focus will be the non-Functional requirements and efficiency. The system will be tested against bigger datasets including 10000 phone numbers, or number plates. This is needed to verify that the system can cope with large amounts of data and will perform well in real life conditions. Also, rule evaluation, and natural language processing pattern detection will be inspected as separate module of the system. This adheres to Objective 1, to confirm that each rule can operate independently, added or removed from the overall system without impact.

For objective 2, the systems ability to detect meaningful patterns in alphanumerical character combinations will be evaluated. This will reveal if the system can match distorted letter combinations to real world words. This includes evaluating performance of approximate string-matching algorithms. This ensures reliability across all modules present in the system and protects the memorability score from distortion.

If temporal difference learning is implemented, adaptability evaluation will ensure that temporal difference learning mechanism work effectively for objective 3. If not implemented, evaluation will focus on predefined rule weights and their influence on suggestions. The weight adjustments will be recorded, to inspect if rule weights adjusted adequately and logically.

3.10 Scalability

This system may be scaled in respect of adding new rules, which can easily be added as separate individual modules with minimal adjustment to overall system.

For future enhancement semantic embedding could be implemented to the system. That would ensure that if the user input certain words, the system would be capable of finding synonyms.

This section outlined the project plan and system workflow, evaluation strategy. The feasibility is given in the next section 4.

4 Feasibility

This section presents the technical and resources feasibility. The system design aims to be lightweight. The modularity enables the scalability and diversion to different kind of system with minimal loss of changes and resources. This project is achievable within the given time scale.

4.1 Technical feasibility

The system design is technically feasible. The system will be built using the Python language, making use of libraries such as Pandas, NumPy. The ranking system is designed in a lightweight way, using micro-agents, which will allow the system to rank and compare a large amount of data in real time. The system is using a modular architecture, keeping data integrity, enabling optimal testing and scalability.

The temporal difference learning algorithm is considered as an optional feature to ensure that the system is responsive to users' preferences without requiring excessive feedback. If implemented, each micro-agent could contribute to the overall memorability score. Such system would be model-free. This approach ensures that the system could be lightweight.

Overall, the algorithms required for such system include rule-based pattern detection, natural language processing methods and temporal difference learning, which can be elegant and straightforward to implement.

4.2 Resource and time feasibility

This project will use open-source software, personal computer, and synthetic datasets; therefore, it will be cost-effective. No external APIs, cloud infrastructure, commercial datasets, or user information required.
The modular design aligns with the semester timeline, allowing the system to be implemented incrementally. See the Gantt chart in section in Appendix.

This section proved, that the proposed system is not only feasible, but also scalable, due to the modular lightweight design. See conclusions in the next section 5.

5 Conclusions

This section is dedicated for conclusions. The first part of conclusions is overviewing motivation, and goals define why such system is useful. And the section Proposed work aims to overview the overall project and possibility for scalability.

5.1 Motivation and goals

This project proposal is an overview of the development of the system that automates the process of finding an easy-to-remember phone numbers or car plates. It takes into consideration users' cognitive preferences, by implementing appropriate computational algorithms. The aim is to determine if a system which is capable of identifying and ranking memorable phone numbers can be developed using predetermined rules.

The proposed system uses rule-based pattern matching, and natural language processing concepts. The cognitive research helped to identify main pattern features and gain insight in people preferences when it comes to memorability. The temporal difference learning may be implemented as optional enhancement to improve the personalization and overcome the rigidness of the predefined importance of rule weights with limited user feedback.

5.2 Proposed work

The design of the project is feasible within the project's timeline and available resources. The Agile methodology used for project planning and management ensures that modular components can be implemented, improved, and tested independently. The Agile methodology ensures that the development process is iterative. This enables efficient development and ease of incremental integration of new functionalities.

In addition, this project is structured with consideration of future scalability. The modular systems design could be upgraded to similar systems which deals with numerical, alphabetical or combination of both data, with minimal changes. The possible improvements with bigger dictionary integration and more advanced methods such as semantic embedding for synonym search.

In conclusion, this project combines cognitive research, and natural language processing concepts to develop a system which is feasible, lightweight, and optimal.

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A. Project Management

For the project management, an Agile approach will be used. It aligns well with the project's modular system design, enabling iterative implementation in short sprints. This type of project management style is flexible and adheres to the need of working with continuous updates, depending on constant feedback. This enables rapid, constant improvements. Each sprint will focus on the development of a handful of operations and testing it.

For tracking the progress, and keeping the meeting minutes with the project supervisor, Notion is used for this project. The OneDrive folder is set up for keeping back up versions of any documentation, diagrams, or code needed for this project. The GitHub repository set, and the access is given to this project supervisor.

Here is the link to project log in Notion https://www.notion.so/Honours-Project-board-283630eded408024a682d3ef169d4f1f?source=copy_link

Semester 1 – Research and design

Weeks 1 – 2 Topic picking and refinement

- Defining research questions
- Identify project scope
- Meeting the project supervisor
- Aims objectives and deliverables

Week 3 – 7 Literature review

- Cognitive memorability research
- Patterns in random number combination
- Similarity metrics
- Adaptive learning systems
- Write findings into a project proposal / research document

Week 8 Requirements and rules

- Functional and Non-functional requirements
- System workflow
- Determine scoring system and rules

Week 9 Methodology and work plan section of project proposal

- Feasibility
- Define stakeholders and use cases
- Methodology section
- Research natural language processing concepts

Week 10 – Work on report

- Feasibility video
- Finalise report

Week 11 Final submissions

- Ensure the project proposal is up to standard
- Incorporate supervisor feedback
- Submit the final project proposal and feasibility video

Semester 2

Week 1 – Setup

- Preparation of datasets that are going to be used for this system.
- Defining test cases for each micro agent.

Weeks 2 – 3 Core rule implementation

- Implement the predefined rules for numeric patterns to the system
- Implement the predefined rules for alphabetic patterns to the system

Week 4 – Ranking and weighted score system

- Implement weighted scores logic
- Ensure compatibility between rules
- Agent score aggregation mechanism

Week 5 – Approximate string matching

- Integrate the approximate string matching
- Tune parameters
- Integrate TD learning

Week 6 – Finalization of the system

- Combine modules into one system

- Ensure that the system works seamlessly with all the components

Week 7 – Testing

- Testing all the components
- Fining and logging the errors
- Parameter adjustments if needed

Week 8 – Evaluation preparation

- Preparation of evaluation datasets
- Preparation of evaluation topics

Week 9 – Evaluation

- Analysis of test and runs
- Comparison of expected performance and actual outcomes
- Create graphs

Week 10 – Finalising dissertation

- Recognise what are strengths and weaknesses of the system
- Finalise dissertation with findings

Week 11 – Submissions

- Format dissertation and ensure that it adheres to standards
- Submit work

A.1. Gantt Chart

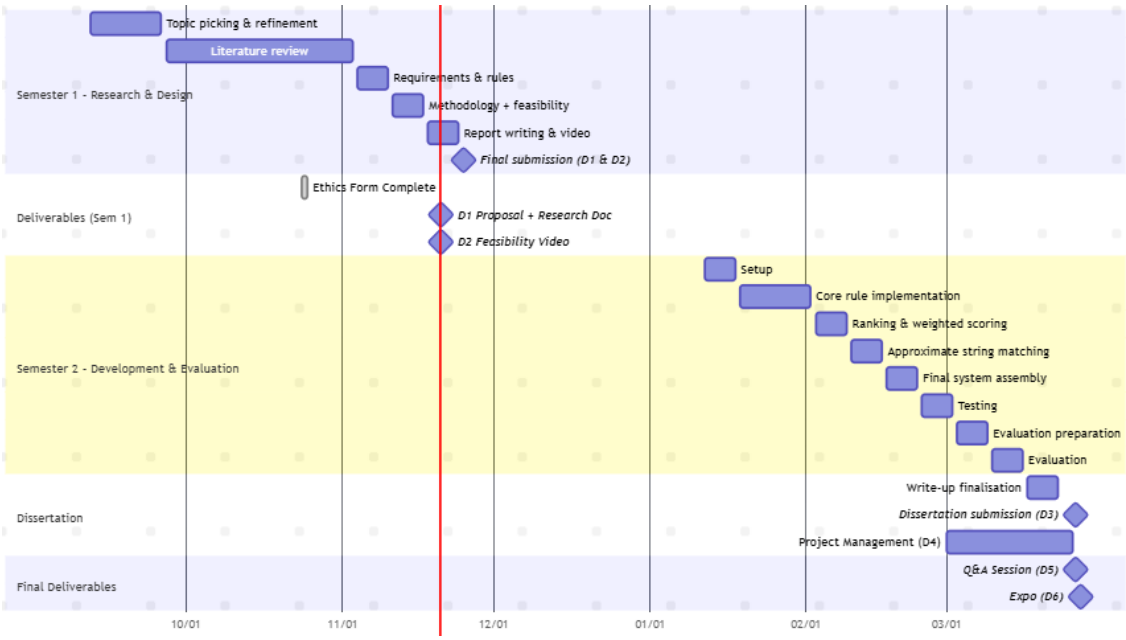

Figure 1: Gantt chart.

A.2. Risk Analysis

Here are some possible risks that may occur during the project's development. Also, on the right side of the table, there are according to mitigations.

	Risks	Likelihood	Impact	Mitigations
1.	Difficulty implementing TD learning.	Medium	High	Ensuring the weight adjustment operations work predictably before integrating user feedback.
2.	Attempting to implement non-essential features before core functionalities are finalized.	High	Medium	Consult the project plan for ensuring that prioritised features are the ones that are being implemented.
3.	Faults in distance logic implementation.	Low	Medium	Testing each distance metric's functionality after implementation.
4.	The ranking operation is too slow for a larger dataset.	Medium	Medium	Identify bottlenecks and prioritize optimization.
5.	Overlapping of the rules.	Low	Medium	Redesign the rules in a way that avoids overlapping.

6.	Dataset quality	Low	Medium	Clean and process the dataset and ensure that missing values are handled.
7.	Struggle to implement all the modules	High	Medium	Integrate incrementally, keeping in mind prioritization.

A.4. Rules

Here are the predetermined rules proposed for the system.
The numeric rules.

	Rule	Explanation	Example
1.	Symmetry	Number sequences that can be read the same from left to right and from right to left. 1 indicates the longest symmetrical sequence and 0 the absence.	2345432, 12321, 1221
2.	Single-digit repetition	Repeated single or collections of digits in a row. It indicates the times number repeats. 10 indicates all numbers repeating, and 0 indicates no repetition present.	222, 344566, 455455
3.	Numerical pattern repetition	Repeated numerical patterns such as pairs or triplets.	233233, 2525
4.	Periodicity	Repeated pattern every few steps.	75767 780781
5.	Sequences	Number sequences. 1 indicates all the single digits, and 0 indicates the absence of sequences in the phone number.	5678, 456
6.	Simple keyboard patterns	Combinations of numbers that make simple rows on the keyboard counter. From max 2 patterns to 0.	123, 456, 789, 741, 852, 963, 159, 753, 357, 951, 8520, 0258.
7.	Unique number count	Unique numbers are present in the phone number. 0 indicates all the unique numbers included in the number, and 1 indicates only one number repeated.	12322123, 45445444
8.	Chunking	Measures the other pattern appearance in certain segments of the number.	555-333, 6969-111

Alphabetic rules

	Rule	Explanation	Example
9.	Letter repetition	Repeated letters in the number plate.	AAA, BB
10.	Letter sequence	Letter sequences appearance.	ABC, EFD
11.	Letter palindromes	Symmetry in letter appearance.	ALA, GYG
12.	Letter symmetry	Visually symmetric letters	IHI, UMU

Alphanumeric rules

	Rule	Explanation	Example
13.	Digit to letter	Digits that look like letters?	H3LLO, C4T
14.	Combination repetition	Combination of a letter and number is repeated.	P1P1, R2R2
15.	Combined symmetry	Symmetry with letter and digits combination.	A22A, B8B
16.	Chunking	Standard plate structure adherence.	ABC 321, AB6 777

Semantic rules

	Rule	Explanation	Example
17.	Match with semantic meaning	Matches the letter combinations to semantically valid words.	DOG, SKY
18.	Match with semantic meaning with digit to letter	Matches the letter and digits like letters combinations to semantically valid words.	B1N, 5KY
19.	Phonetic formation	Matches certain numbers and letters to their phonetic pronunciation in combination with letters.	B4, GR8

A.5. Diagrams

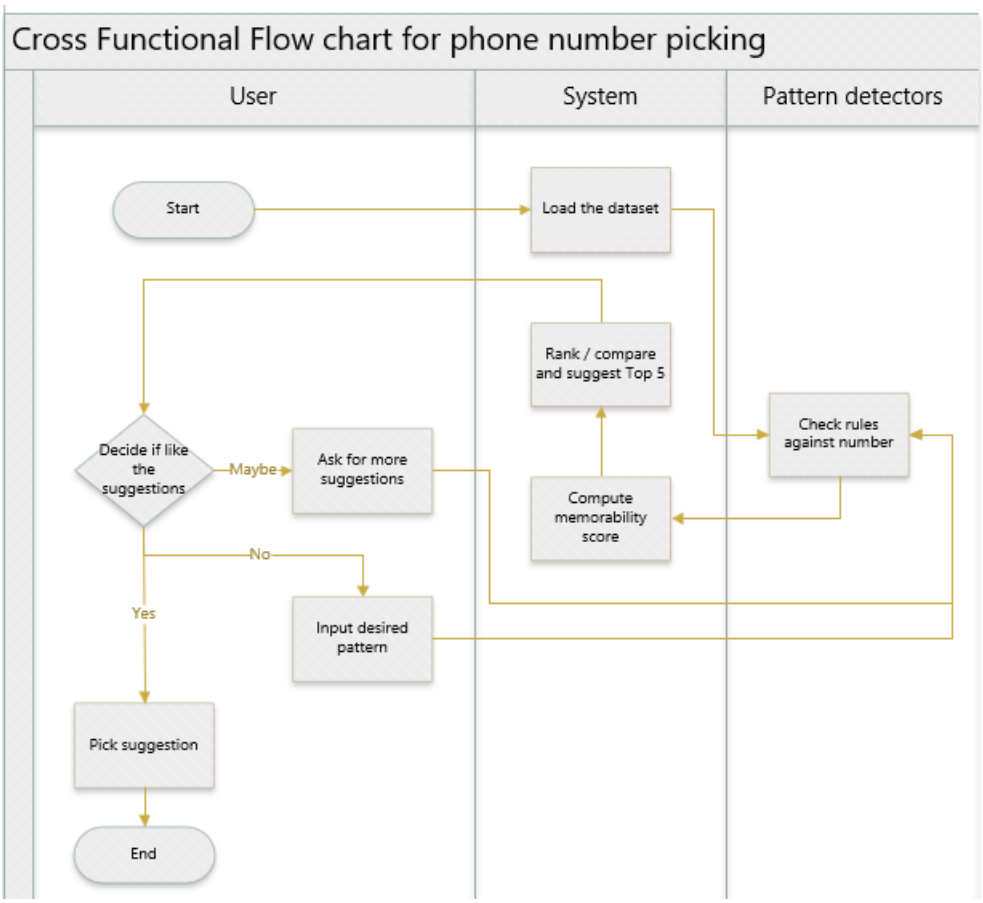

Figure 2: Cross functional flowchart for number picking.

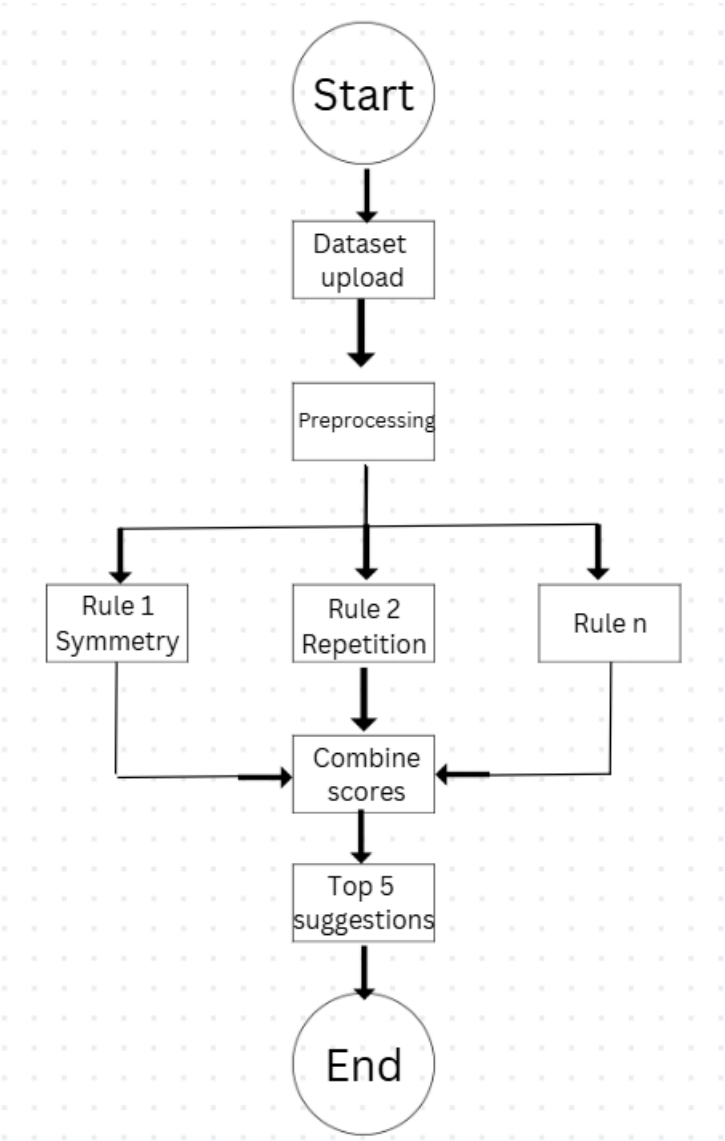

Figure 3: Box diagram.

A.1. Ples

B. Generative AI

No generative AI was used in the making of this report. The only editing assistance that I used in this project was the built-in Microsoft Word processor for spelling mistakes.